

Group introductions

o Introduce yourself

- Provide name, major, and year
- Why did you decide to take this course?
- What is one thing you hope to get out of it?

o Myself

- James Fitzgerald
- Associate Professor of Neurobiology
- Secondary appointments in Physics and Astronomy, Engineering Sciences and Applied Mathematics
- My research is theoretical neuroscience
- Degrees in physics and math, bio research in college, neuroscience research in graduate school
- Neuro is interdisciplinary, theory and computation are increasingly central and should be accessibly taught.
- My goal is to make the subject interesting and accessible to the widest range of motivated students.

Group Discussion:

"What is computational neuroscience?"

- Using computers as a means to study the brain
 - Data analysis
 - Machine learning & neural networks
 - Modeling the nervous system
 - Simulating models
- } "Neural data science"
- Studying important computational functions of the brain
 - How do sensory systems extract meaningful signals from raw sensory data?
 - How does the brain control the body?
 - How does the brain store & retrieve memories?
 - How does this all come together to form an intelligent system?

"How is computational neuroscience relevant?"

- Engineering
 - Machine learning & neural networks
 - Machine vision
 - Robotics
 - AI
- Medicine
 - Brain-computer interfaces
 - Computational psychiatry

Syllabus overview:

- Go over the whole thing. Questions?
- Make sure to discuss:
 - Course numbers
 - TAs
 - What is section?
 - Office hour choices
 - Getting to our wing of Cook
 - What's happening this week?
 - No daily questions today or tomorrow
 - Assigned working groups
 - Timeline for selecting paper (second or third week)
 - Underline desire for feedback
 - Zoom policy

Mathematics and computation in course:

- Discuss the pre-reqs for the course. Why?
- How are math and programming useful?
 - Stating and framing the fundamental concepts
 - Deriving and understanding results from concepts.
 - Programming is a key skill for both experimenters and theorists.
 - MATLAB is widely used in field and in NU courses. Python is growing in popularity.
- Take survey.
- Discuss survey results.
- Explain flipped classroom logic and logistics.
- Share Flipped classroom exercises. Important to:
 - read it
 - get MATLAB up and running
 - confirm your understanding
 - flag your questions